

Nguyen Manh Hiep

AI-Native Full-Stack Developer

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Summary

AI-native full-stack developer who ships complete products, not prototypes. I drive the whole lifecycle (requirements, architecture, frontend, backend, database, tests, deployment) with an AI-first workflow — Claude Code, Cursor, ChatGPT, Gemini and Copilot as the primary build engine, not an autocomplete helper. Latest proof is **SenAI**, a real-time Vietnamese–English meeting interpreter — **~29k LOC in 336 commits over 3 weeks**, led across frontend, backend and infrastructure, live in production at app.senai.online, **3rd Prize + Vietnam AI Stars at the Vietnam AI Innovation Challenge 2026 (AI Singapore)**.

Career Objective

Own products end-to-end as an AI-Native Full-Stack Developer — business document to deployed MVP in weeks — and push the team's AI development workflow (prompt & context engineering, MCP, agent guardrails) into something measurably faster and safer than hand-written delivery. Next, I am deepening scalable real-time architecture and production LLM/ML integration (routing, evaluation, cost and latency budgets), aiming to lead a small cross-functional team that turns AI-native delivery into a repeatable engineering standard rather than a one-off experiment.

Selected Achievements

- **3rd Prize + Vietnam AI Stars**, Vietnam AI Innovation Challenge 2026 (AI Singapore), for SenAI — a real-time Vietnamese–English meeting interpreter shipped to public production.
- **4 production systems delivered end-to-end** in ~12 months (SenAI, Notarex, EasySale, AKF Farm) — **~155,000 LOC across 859 commits** on the three repositories I own.
- **Blank repo to live production in 3 weeks** on SenAI (336 commits) — full stack, database, security and deployment, with the AI workflow doing the bulk of the typing.
- **1,000+ automated / UAT test cases authored** across the four products (418 Notarex, 347 EasySale, 147 SenAI real non-mocked, 132 AKF Farm) — every AI-generated feature ships behind a real test.
- **Sub-400 ms live translation** (~75 ms ASR + ~295 ms MT measured end-to-end) with zero errors up to concurrency 8, and runtime provider switching with no restart and no downtime.

AI-Native Development Workflow

Toolchain: Claude Code (primary agent, subagents, custom skills), Cursor, ChatGPT, Gemini, GitHub Copilot; MCP servers (GitNexus code intelligence, browser automation) wired into the coding loop

Context engineering: Repo-level agent context (CLAUDE.md / AGENTS.md) pinning architecture, conventions, protocol source-of-truth and branch policy, so agents produce merge-ready code without re-briefing

Spec-driven loop: Brainstorm → versioned design spec (docs/specs/) → implementation plan (docs/plans/) → AI implementation → real test run; one canonical spec per system, never parallel versions

AI guardrails: Every AI-written feature lands behind real no-mock UAT (live API/WebSocket/LiveKit calls), call-graph impact analysis before editing a symbol, and CI on every push

Projects

SenAI — Real-time Vietnamese–English Meeting Interpreter (app.senai.online)

2026

AI-native platform for live bilingual meetings — per-speaker ASR, translation and voice cloning, automatic minutes, on-prem or commercial model routing. Built for the Vietnam AI Innovation Challenge 2026 (AI Singapore) — **3rd Prize + Vietnam AI Stars**. Lead developer, 298 of 336 commits across frontend, backend and infrastructure.

- **AI-native delivery:** ~29,000 LOC in 336 commits over 3 weeks, built with Claude Code on a spec-driven loop (design spec → plan → agent implementation → real UAT), from a blank repo to a public production deployment.
- **Full-stack ownership:** Next.js 16 / React 19 / Tailwind v4 app with bilingual VI–EN i18n (~370 keys); layered FastAPI backend (routers / engine / store / model router); PostgreSQL 16 + pgvector for meetings, transcripts, glossaries and 1024-dim embeddings, auto-migrated on boot.
- **Real-time AI pipeline:** one independent ASR+MT pipeline per LiveKit mic track (no shared buffers), 2-lane fast→refine subtitles and a streaming *Cabin* mode that translates while the speaker is still talking. Measured **~75 ms ASR + ~295 ms MT (<400 ms end-to-end)**, zero errors up to concurrency 8.
- **Pluggable Model Router:** one capability contract (asr/mt_fast/mt_refine/tts/llm) behind self-hosted models (Qwen3, F5-TTS on vLLM) or commercial APIs (Google, Deepgram, OpenAI, ElevenLabs); pipeline, provider and keys switch at runtime from an admin UI with auto-fallback.

- **RAG & prompt engineering:** glossaries and uploaded documents chunked into pgvector embeddings, LLM term extraction, ASR phonetic correction, bilingual LLM summaries; prompt-injection defenses covered by a pentest suite.
- **Auth, security & integrations:** Google OAuth + email sign-in, JWT access/refresh rotation, rate limiting, per-meeting RBAC and audit logs; an on-prem-only mode blocking every cloud call for data residency; a public AI Gateway (/ai/*) with API keys; Amazon SES PDF minutes.
- **Deployment & QA:** VPS on Nginx + systemd + TLS with pull-based CI/CD, GitHub Actions CI on every push, Docker for LiveKit and PostgreSQL; **37 UAT suites / 147 real (non-mocked) tests.**

Notarex — Notary Document OCR & Digitization

2026

Digitizes notarial records — AI OCR from scanned papers, online editing, automatic Word generation. Role — full-stack, AI orchestration and infrastructure.

- Built configurable form templates, the AI OCR extraction flow and an edit/save/download workflow through JWT-secured OnlyOffice, with automatic DOCX generation via docxtpl.
- Designed a 3-role permission model (admin / director / employee) scoped per notary office for tenant isolation.
- Built a Celery + Redis job queue (ZSET pending queue, atomic active-counter, heartbeat reconciliation) that detects and recovers zombie OCR jobs under load; MinIO (S3) storage, Dockerized behind Nginx with GitHub Actions CI/CD.
- Scale of delivery — **387 commits and ~55,000 LOC over 3.5 months**, covered by **43 test modules / 418 automated test cases.**

EasySale — Sales Lead Management & Automation

2026

Automation platform that collects leads and nurtures them without manual follow-up.

- Built a Chrome Extension that scrapes business leads from Google Maps straight into the system.
- Integrated the Zalo OA API for automated care messages and SendGrid for email campaigns.
- Delivered separate admin and customer portals behind a unified login, deployed with Docker + Nginx.
- Scale of delivery — **136 commits and ~71,000 LOC**, covered by **19 test modules / 347 automated test cases.**

AKF Farm — Agricultural Production Management (Web Admin + Mobile PWA)

2026

Production management for agriculture, built to stay reliable in low-connectivity fields.

- Built a headless Frappe v15 backend with an event-driven task engine (step / frequency / prerequisite), a worst-of status model, load-balanced assignment, REST API and 2-role auth — **verified by 132 passing tests.**
- Anti-fraud completion photos (in-app camera, GPS, watermark) plus an idempotent offline queue so field data is never lost or duplicated.
- Delivered a satellite heatmap (Leaflet + Esri), work calendar, notifications and step-by-step SOPs.

Education

Hanoi University of Science and Technology (HUST), BSc in Computer Science

2021 – 2026

- Coursework applied directly in production work — data structures & algorithms, databases, software engineering, computer networks, operating systems, machine learning.

Skills

AI-assisted development: Claude Code (agents, skills, hooks), Cursor, ChatGPT, Gemini, GitHub Copilot, MCP servers, prompt & context engineering, spec-driven AI workflows, RAG, LLM orchestration, vLLM self-hosting

Frontend: React 19, Next.js 16/14, TypeScript, Tailwind v4, Ant Design, Zustand, TanStack Query, PWA / offline-first, i18n

Backend: FastAPI, Node.js, NestJS / ExpressJS, Spring Boot, Python, Java, REST, WebSocket, WebRTC (LiveKit), Celery, Redis, JWT / OAuth, RBAC

Data: PostgreSQL 16 (+ pgvector), MySQL, MariaDB, SQLAlchemy 2.0 async, schema migrations, vector search & embeddings, MinIO (S3)

DevOps & Cloud: Docker & Docker Compose, Nginx, systemd, TLS, VPS deployment, GitHub Actions CI/CD, Cloudflare, Amazon SES

Engineering practice: Systems thinking, software architecture, SOLID, OOP, design patterns, real (non-mocked) integration & security testing, requirements analysis, SRS / Use Case / ERD

Languages: Vietnamese (native), English (technical reading, writing and documentation)